

HDOC – Experiment 24

Question

Write a C program that displays the following menu and uses a switch statement to print the “Increasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameter less function with no return value.

❖ Problem understanding

The program displays a menu containing five increasing patterns. The user selects one option using a switch statement.

Each pattern is implemented using:

- A separate parameter less void function
- Nested loops
- No arrays
- Validation for the number of rows
- A menu that continues until the user chooses Exit

Menu

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit

❖ Algorithm

Main Program Algorithm

1. Start.
2. Declare integer variable choice.
3. Display the menu.
4. Input choice.
5. Use a switch statement:
 - If choice = 1, call rightTriangle().
 - If choice = 2, call rowNumberTriangle().
 - If choice = 3, call naturalNumberPattern().
 - If choice = 4, call consecutiveNumberTriangle().
 - If choice = 5, call alphabetPattern().
 - If choice = 6, display an exit message.
 - Otherwise, display "Invalid choice."
6. Repeat Steps 3–5 until the user selects 6.
7. Stop.

Algorithm 1: Increasing Right-Angled Triangle

1. Start the function.
2. Input number of rows n.
3. If $n \leq 0$, display an error message and return.
4. Repeat i from 1 to n.
5. For each row, repeat j from 1 to i.
6. Print *.
7. After completing each row, print a newline.
8. Return to the main program.

Algorithm 2: Increasing Row-Number Triangle

1. Start the function.
2. Input number of rows n .
3. If $n \leq 0$, display an error message and return.
4. Repeat i from 1 to n .
5. Repeat j from 1 to i .
6. Print the value of i .
7. Print a newline after each row.
8. Return to the main program.

Algorithm 3: Increasing Natural-Number Pattern

1. Start the function.
2. Input number of rows n .
3. If $n \leq 0$, display an error message and return.
4. Repeat i from 1 to n .
5. Repeat j from 1 to i .
6. Print j .
7. Print a newline after each row.
8. Return to the main program.

Algorithm 4: Consecutive-Number Triangle

1. Start the function.
2. Input number of rows n .
3. If $n \leq 0$, display an error message and return.
4. Initialize $num = 1$.
5. Repeat i from 1 to n .

6. Repeat j from 1 to i.
7. Print num.
8. Increment num by 1.
9. Print a newline after each row.
10. Return to the main program.

Algorithm 5: Increasing Alphabet Pattern

1. Start the function.
2. Input number of rows n.
3. Check whether n is between 1 and 26.
4. If not, display an appropriate error message and return.
5. Repeat i from 1 to n.
6. Repeat j from 1 to i.
7. Print the uppercase letter obtained using 'A' + j - 1.
8. Print a newline after each row.
9. Return to the main program.

❖ Test case table

Test Case	Menu Choice	Input n	Expected Output	Actual Output	Result
1	1	4	4-row increasing * triangle	Same as expected	Pass
2	2	4	Rows 1, 2 2, 3 3 3, 4 4 4 4	Same as expected	Pass
3	3	4	Natural-number triangle up to row 4	Same as expected	Pass
4	4	4	Consecutive numbers from 1 to 10 in triangular form	Same as expected	Pass
5	5	4	Alphabet pattern from A to D	Same as expected	Pass
6	1	1	*	Same as expected	Pass
7	3	0	Error: rows must be positive	Same as expected	Pass
8	4	-3	Error: rows must be positive	Same as expected	Pass
9	5	26	Valid alphabet pattern ending with Z	Same as expected	Pass
10	5	27	Error: rows must be 1 to 26	Same as expected	Pass
11	9	—	Invalid menu choice	Same as expected	Pass
12	6	—	Program terminates	Same as expected	Pass

❖ Program code in C

```
#include <stdio.h>
```

```
/* Function declarations */
```

```
void rightTriangle(void);

void rowNumberTriangle(void);

void naturalNumberPattern(void);

void consecutiveNumberTriangle(void);

void alphabetPattern(void);


int main(void)
{
    int choice;


    do
    {
        printf("\n===== INCREASING PATTERNS MENU =====\n");
        printf("1. Increasing right-angled triangle\n");
        printf("2. Increasing row-number triangle\n");
        printf("3. Increasing natural-number pattern\n");
        printf("4. Consecutive-number triangle\n");
        printf("5. Increasing alphabet pattern\n");
        printf("6. Exit\n");


        printf("Enter your choice: ");
        scanf("%d", &choice);


        switch (choice)
        {
            case 1:
                rightTriangle();
                break;
```

case 2:

 rowNumberTriangle();

 break;

case 3:

 naturalNumberPattern();

 break;

case 4:

 consecutiveNumberTriangle();

 break;

case 5:

 alphabetPattern();

 break;

case 6:

 printf("Exiting the program...\n");

 break;

default:

 printf("Invalid choice. Please enter a choice from 1 to 6.\n");

}

} while (choice != 6);

return 0;

```
}
```

```
/* 1. Increasing right-angled triangle */
```

```
void rightTriangle(void)
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter the number of rows: ");
```

```
    scanf("%d", &n);
```

```
    if (n <= 0)
```

```
    {
```

```
        printf("Error: Number of rows must be positive.\n");
```

```
        return;
```

```
    }
```

```
    for (i = 1; i <= n; i++)
```

```
    {
```

```
        for (j = 1; j <= i; j++)
```

```
        {
```

```
            printf("* ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
/* 2. Increasing row-number triangle */
```

```
void rowNumberTriangle(void)
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter the number of rows: ");
```

```
    scanf("%d", &n);
```

```
    if (n <= 0)
```

```
    {
```

```
        printf("Error: Number of rows must be positive.\n");
```

```
        return;
```

```
    }
```

```
    for (i = 1; i <= n; i++)
```

```
    {
```

```
        for (j = 1; j <= i; j++)
```

```
        {
```

```
            printf("%d ", i);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
/* 3. Increasing natural-number pattern */
```

```
void naturalNumberPattern(void)
```

```
{
```

```
int n, i, j;
```

```
printf("Enter the number of rows: ");
```

```
scanf("%d", &n);
```

```
if (n <= 0)
```

```
{
```

```
    printf("Error: Number of rows must be positive.\n");
```

```
    return;
```

```
}
```

```
for (i = 1; i <= n; i++)
```

```
{
```

```
    for (j = 1; j <= i; j++)
```

```
    {
```

```
        printf("%d ", j);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
}
```

```
/* 4. Consecutive-number triangle */
```

```
void consecutiveNumberTriangle(void)
```

```
{
```

```
    int n, i, j, num = 1;
```

```
    printf("Enter the number of rows: ");
```

```
scanf("%d", &n);
```

```
if (n <= 0)
```

```
{
```

```
    printf("Error: Number of rows must be positive.\n");
```

```
    return;
```

```
}
```

```
for (i = 1; i <= n; i++)
```

```
{
```

```
    for (j = 1; j <= i; j++)
```

```
    {
```

```
        printf("%d ", num);
```

```
        num++;
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
}
```

```
/* 5. Increasing alphabet pattern */
```

```
void alphabetPattern(void)
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter the number of rows (1-26): ");
```

```
    scanf("%d", &n);
```

```
if (n < 1 || n > 26)
{
    printf("Error: Number of rows must be between 1 and 26.\n");
    return;
}

for (i = 1; i <= n; i++)
{
    for (j = 1; j <= i; j++)
    {
        printf("%c ", 'A' + j - 1);
    }
    printf("\n");
}
}
```

❖ Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/exp24]
└─$ vi increasing_patterns.c

(kali㉿kali)-[~/Desktop/exp24]
└─$ gcc increasing_patterns.c -o increasing_patterns

(kali㉿kali)-[~/Desktop/exp24]
└─$ gcc -Wall -Wextra increasing_patterns.c -o increasing_patterns

(kali㉿kali)-[~/Desktop/exp24]
└─$ ./increasing_patterns

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 1
Enter the number of rows: 4
*
* *
* * *
* * * *

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 2
Enter the number of rows: 4
1
2 2
3 3 3
```

```
1
2 2
3 3 3
4 4 4 4

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 3
Enter the number of rows: 4
1
1 2
1 2 3
1 2 3 4

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 4
Enter the number of rows: 4
1
2 3
4 5 6
7 8 9 10

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
```

```

6. Exit
Enter your choice: 5
Enter the number of rows (1-26): 4
A
A B
A B C
A B C D

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 1
Enter the number of rows: 1
*

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 3
Enter the number of rows: 0
Error: Number of rows must be positive.

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 4
Enter the number of rows: -3

```

```

Error: Number of rows must be positive.

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 5
Enter the number of rows (1-26): 26
A
A B
A B C
A B C D
A B C D E
A B C D E F
A B C D E F G
A B C D E F G H
A B C D E F G H I
A B C D E F G H I J
A B C D E F G H I J K
A B C D E F G H I J K L
A B C D E F G H I J K L M
A B C D E F G H I J K L M N
A B C D E F G H I J K L M N O
A B C D E F G H I J K L M N O P
A B C D E F G H I J K L M N O P Q
A B C D E F G H I J K L M N O P Q R
A B C D E F G H I J K L M N O P Q R S
A B C D E F G H I J K L M N O P Q R S T
A B C D E F G H I J K L M N O P Q R S T U
A B C D E F G H I J K L M N O P Q R S T U V
A B C D E F G H I J K L M N O P Q R S T U V W
A B C D E F G H I J K L M N O P Q R S T U V W X
A B C D E F G H I J K L M N O P Q R S T U V W X Y
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

```

```
===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 5
Enter the number of rows (1-26): 27
Error: Number of rows must be between 1 and 26.

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 9
Invalid choice. Please enter a choice from 1 to 6.

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 6
Exiting the program...

(kali@kali) - [~/Desktop/exp24]
$
```